Parameter estimation

Home assignment 1 A – model structure selection

Consider a set of ARX and OE 2nd order models parameterized by polynomials

$$a(d) = 1 - 1.5\,d + 0.6\,d^2,$$

$$b(d) = 0.025*(1 + 2\,d + d^2).$$

1. Generate input-output data for both model structures and two different excitation signals (PRBS with low frequency and white noise spectra). Select the noise covariance σ_e^2 to achieve comparable signal/noise ratios - noise amplitude (2..3 σ_y) appx 50% of signal amplitude - for both model structures.

2. Analyze the convergence of recursive ARX/ARMAX/OE parameter estimation using linear/pseudolinear regression methods. Apply each model structure to all noise models. Demonstrate the impact of correct structure selection and excitation signal on parameter convergence.

3. In the case of pseudo linear regression (ARMAX and OE model), demonstrate the impact of replacement of the output prediction / prediction error used in the regressor

   $$y(t|t-1) = z^T(t)\Theta(t|t-1), \qquad \varepsilon(t|t-1) = y(t) - y(t|t-1)$$

   by filtered values

   $$y(t|t) = z^T(t)\Theta(t|t), \qquad \varepsilon(t|t) = y(t) - y(t|t),$$

   and use of iterative estimation (estimated C-parameters used as initial estimate for the next iteration).

4. Evaluate the structure selection strategy “ARMAX first” (decision of the model structure based on the estimated c(d) parameters – if estimated c(d) = 1, select ARX, if estimated c(d) = a(d), select OE.

5. Conclude with a summary of “lessons learned”.

Deliverables: report (**max 16 pages, naming convention HW1A_TeamX.pdf file**) + Matlab scripts (zip file) on demand

Due date: November 13, 2026